How to *Fail* at Destroying a Black Hole

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Abstract: We attempt to destroy a black hole in the most plausible way. The attempt fails.

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1 Throw a stone first

Throw a stone upward. If it is slow, it returns. Throw it faster and it travels farther before returning. Newtonian mechanics compresses the contest between kinetic and gravitational energy into one number, the escape velocity

$$v_{\mathrm{esc}} = \sqrt{\frac{2GM}{R}}.$$

In 1783, Michell followed this formula into deep territory. If a star were so compact that $2GM/R > c^2$, then even corpuscular light would fail to escape. His paper appeared the following year [1]. Michell's object was of course not a relativistic black hole. It had a material surface, light was treated in a Newtonian theory and nothing resembling the modern definition of an event horizon had yet been invented. Still, the thought was irresistible, the thought that gravity might hide an object not because the object emits nothing, but because what it emits cannot get away. General relativity changes the verb. A black hole does not merely pull light back. It arranges spacetime so that every future directed causal route from an interior region remains interior. The importance is but a causal boundary.

That tells us what "destroying a black hole" must mean. It cannot mean making the hole radiate, extracting some of its rotational energy, merging it with another hole, erasing its shadow or even removing its singularity. The proposed destruction is perverse (liking to behave in a way that is not acceptable or reasonable or that most people think is wrong). We try to remove the horizon while leaving a singular region behind. We aren't smashing the safe, we are just stealing its door (otherwise, there'd be no bank). This is a test of *weak cosmic censorship*.

Penrose's singularity theorem showed that gravitational collapse can force causal geodesic incompleteness under broad geometric assumptions [2]. A theorem that predicts singular behaviour is not yet a theorem that hides it. Penrose's later cosmic censorship proposal gave the missing hope: singularities produced by generic, physically reasonable collapse should not be visible from infinity [3]. The word *generic* has carried a heroic amount of unfinished mathematics ever since! The best attempted burglary begins with a perfectly respectable black hole, adds ordinary matter, and asks whether the final spacetime can be forced outside the black hole family. Robert Wald posed exactly this sort of Gedankenexperiment (German for what Einstein did best, thought experiments) in 1974 [4].

We shall need three ledgers, and we ought to keep them open throughout.

1. The *geometric ledger*: does the proposed final set (M, J, Q) admit a Kerr–Newman horizon?

2. The *admission ledger*: can matter carrying (E, L, q) actually cross the existing horizon?

3. The *perturbative ledger*: have we retained every term of the order on which the verdict depends?

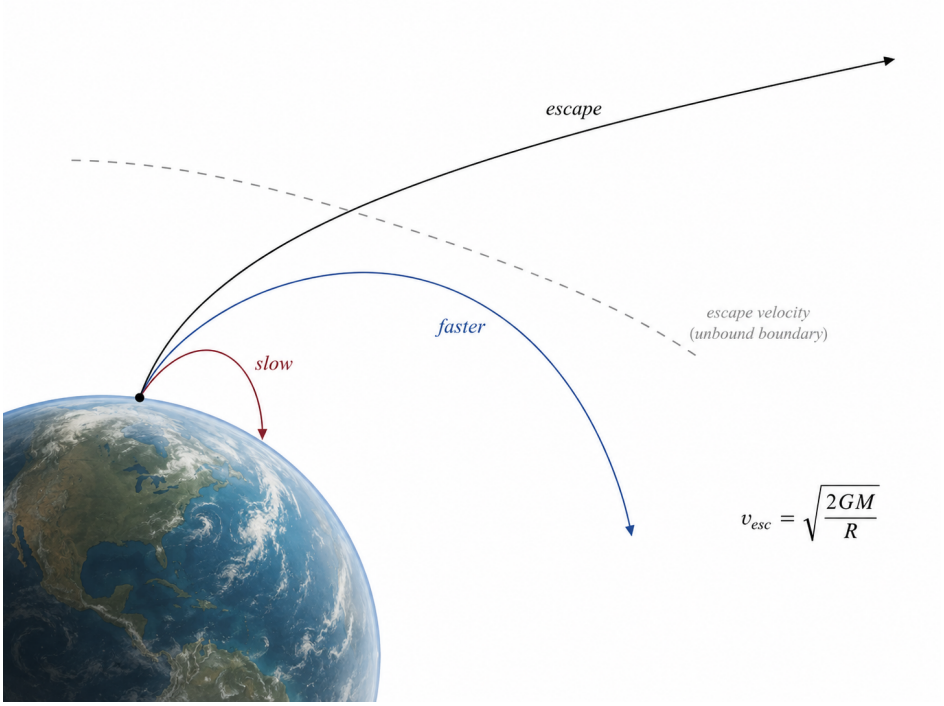


Figure 1: Escaping the Earth

2 What, exactly, is the door?

Let $(\mathcal{M}, g_{ab})$ be an asymptotically flat spacetime with future null infinity $\mathscr{I}^+$. Its black hole region is

$$\mathcal{B} = \mathcal{M} \setminus J^-(\mathscr{I}^+),$$

and its future event horizon is

$$\mathcal{H}^+ = \partial J^-(\mathscr{I}^+).$$

An event lies in $\mathcal{B}$ when no future directed causal curve from it reaches $\mathscr{I}^+$. This definition is global: locating the exact event horizon requires knowledge of the entire future development of the spacetime. The horizon is not a substance, not a locally painted shell, and not generically the place where a convenient metric component becomes infinite.

Why, then, does nearly every black hole calculation begin by setting some function of r equal to zero? Because stationarity gives us extraordinary shortcuts. Within an exact stationary family, a Killing horizon can be located algebraically. The shortcut is exact inside that family and it is not (and must not be) a replacement for the global definition in an arbitrary dynamical spacetime. This will matter at the end. Showing that a proposed final (M, J, Q) is superextremal proves that no stationary Kerr–Newman black hole has those charges. It does not, by itself, solve the complete nonlinear evolution and exhibit a null ray escaping from a singularity. The stationary inequality is our target condition. So smart people have started to use these geometrized units $G = c = 4\pi\varepsilon_0 = 1$. Then M, Q, and a have dimensions of length, J has dimensions of length squared, and $a = J/M$. The quantity a is convenient but not independently conserved: when M and J change, a changes with them.

In Boyer–Lindquist coordinates, the Kerr–Newman metric can be written as

$$\mathrm{d}s^2 = -\frac{\Delta}{\Sigma}\left(\mathrm{d}t - a\sin^2\theta\,\mathrm{d}\phi\right)^2 + \frac{\sin^2\theta}{\Sigma}\left[(r^2+a^2)\mathrm{d}\phi - a\,\mathrm{d}t\right]^2 + \frac{\Sigma}{\Delta}\,\mathrm{d}r^2 + \Sigma\,\mathrm{d}\theta^2$$

where $\Sigma = r^2 + a^2\cos^2\theta$ and $\Delta = r^2 - 2Mr + a^2 + Q^2$. A compatible electromagnetic potential is $A_a\mathrm{d}x^a = (-Qr/\Sigma)(\mathrm{d}t - a\sin^2\theta\,\mathrm{d}\phi)$. Kerr found the rotating vacuum geometry in 1963 [5] and Newman and collaborators added electric charge soon afterward [6]. The finished metric looks forbidding, but the part relevant to our first ledger is just a quadratic polynomial (let's gooo). Since $g^{rr} = \Delta/\Sigma$, the stationary candidate horizons occur at the roots of Δ, and these are

$$r_\pm = M \pm \sqrt{M^2 - a^2 - Q^2}.$$

The outer root r_+ is the event horizon of the stationary black hole while r_- is the inner, or Cauchy, horizon. Completing the square makes

$$\Delta = (r - M)^2 - (M^2 - a^2 - Q^2).$$

$$\begin{aligned} M^2 > a^2 + Q^2 &\implies \text{two distinct real roots.} \\ M^2 = a^2 + Q^2 &\implies \text{one degenerate root.} \\ M^2 < a^2 + Q^2 &\implies \text{no real root.} \end{aligned}$$

The curvature singularity occurs instead at $\Sigma = 0$, namely $r = 0$ and $\theta = \pi/2$. The divergence at $\Delta = 0$ in Boyer–Lindquist coordinates is removable by horizon penetrating coordinates but the divergence at $\Sigma = 0$ is not. Passing from subextremal to superextremal Kerr–Newman therefore removes the stationary horizon roots without removing the ring singularity. That is the intended crime. Because $a = J/M$, the clean polynomial measure of extremality is

$$\mathcal{F}(M, J, Q) = M^4 - M^2Q^2 - J^2.$$

We shall not define $\mathcal{F}$ and forget it. It is the geometric ledger.

$$\begin{aligned} \mathcal{F} > 0 &\implies \text{subextremal.} \\ \mathcal{F} = 0 &\implies \text{extremal.} \\ \mathcal{F} < 0 &\implies \text{superextremal.} \end{aligned}$$

Equivalently, we can define

$$f(M, J, Q) = M^2 - Q^2 - \frac{J^2}{M^2} = \frac{\mathcal{F}}{M^2}$$

so f has dimensions of length squared and the same sign as $\mathcal{F}$. A useful nonextremality scale would be

$$\epsilon = \frac{\sqrt{M^2 - a^2 - Q^2}}{M} \tag{1}$$

so that $f = M^2\epsilon^2$ and near extremality, $0 < \epsilon \ll 1$. Notice already: the distance from extremality is quadratic in ϵ. For a picture, we can introduce $\chi = J/M^2$ and $\zeta = Q/M$. At fixed positive M, stationary Kerr–Newman black holes occupy $\chi^2 + \zeta^2 \leq 1$. The Schwarzschild hole is at the centre, Kerr and Reissner–Nordström occupy the two axes while extremal holes

lie on the boundary circle. The proposed experiment tries to move a point from the disk to its exterior. Now we define

$$A_{\mathrm{H}} = 4\pi(r_+^2 + a^2), \qquad \Omega_{\mathrm{H}} = \frac{a}{r_+^2 + a^2}, \qquad \Phi_{\mathrm{H}} = \frac{Qr_+}{r_+^2 + a^2}, \tag{2}$$

$$\kappa = \frac{r_+ - r_-}{2(r_+^2 + a^2)} = \frac{M\epsilon}{r_+^2 + a^2}.$$

These are, respectively, the horizon area, angular velocity, electric potential relative to infinity, and surface gravity. At extremality $r_+ = r_- = M$ and $\kappa = 0$, but $A_{\mathrm{ext}} = 4\pi(M^2 + a^2)$ is finite. Extremality is the moment the two stationary horizon roots coalesce. Destruction requires crossing the boundary, not merely reaching it.

3 The bribe

Suppose the initial hole has (M, J, Q), and an incoming body or field carries energy, angular momentum, and charge (E, L, q). The naive test body ledger reads $M_{\mathrm{f}} = M + E$, $J_{\mathrm{f}} = J + L$ and $Q_{\mathrm{f}} = Q + q$. The proposed final state is superextremal if

$$(M + E)^4 - (M + E)^2(Q + q)^2 - (J + L)^2 < 0. \tag{3}$$

That is the destruction condition. It asks for a large L/E, a large q/E or a cunning mixture of both. There is an immediate obstacle. The same body must be captured. A particle with a heroic charge to energy ratio may be electrostatically repelled. A particle with a heroic angular momentum to energy ratio may meet a centrifugal turning point. A wave selected for its generous angular momentum may be superradiantly reflected. The projectile must be simultaneously dangerous enough to win the geometric ledger and tame enough to pass the admission ledger. The horizon now reveals itself as a *very strict cashier*. Let t^a and φ^a denote the stationary and axial Killing fields. The future horizon is generated by $\xi^a = t^a + \Omega_{\mathrm{H}}\varphi^a$. For a particle of rest mass μ, charge q, four velocity u^a, and canonical momentum $P_a = \mu u_a + qA_a$, the conserved energy and axial angular momentum are $E = -t^a P_a$ and $L = \varphi^a P_a$. On the horizon, $\Phi_{\mathrm{H}} = -\xi^a A_a$. Therefore

$$E - \Omega_{\mathrm{H}} L - \Phi_{\mathrm{H}} q = -\mu u_a \xi^a. \tag{4}$$

Both u^a and ξ^a are future directed, with u^a timelike and ξ^a null on $\mathcal{H}^+$, hence $u_a \xi^a \leq 0$. Any particle crossing the future horizon must obey

$$E - \Omega_{\mathrm{H}} L - \Phi_{\mathrm{H}} q \geq 0. \tag{5}$$

Equality is an ideal limiting process in which the particle arrives with vanishing kinetic energy relative to the horizon generator. For extended matter, the corresponding first order flux law is

$$\delta M - \Omega_{\mathrm{H}} \delta J - \Phi_{\mathrm{H}} \delta Q \geq 0, \tag{6}$$

provided the non electromagnetic stress tensor satisfies the null energy condition $T^{\mathrm{matter}}_{ab} k^a k^b \geq 0$ for every null k^a and the appropriate flux and asymptotic assumptions hold. The left hand

side is the flux of Killing energy associated with ξ^a through the horizon. Equation (6) says that charge and angular momentum are not admitted free of energy.

$$\delta M \geq \Omega_{\mathrm{H}} \delta J + \Phi_{\mathrm{H}} \delta Q$$

is the clear demand. Now for a charged field mode $\Psi \sim e^{-i\omega t + im\phi}$, the frequency seen by the horizon generator is

$$\widetilde{\omega} = \omega - m\Omega_{\mathrm{H}} - q\Phi_{\mathrm{H}}. \tag{7}$$

Ordinary absorption corresponds to nonnegative horizon energy. For bosonic modes with

$$0 < \omega < m\Omega_{\mathrm{H}} + q\Phi_{\mathrm{H}},$$

the reflected wave can emerge amplified, the superradiant regime. A wave carrying too much angular momentum or charge per unit asymptotic energy is not a brilliant new attack. It has asked the horizon to sell below its own price, and the horizon responds by sending energy back. Particle capture and superradiance are not unrelated. They are the same contraction with the same generator ξ^a but one is in trajectories and one in the modes.

4 Wald's two perfect failures

The best attack begins with an exactly extremal hole. This sounds sensible, if the system already lies on $\mathcal{F} = 0$, the smallest outward nudge should make $\mathcal{F} < 0$. Instead, exact extremality kind of actually makes the contradiction sharp.

4.1 Attempt I: Overcharge an extremal Reissner–Nordström hole

Let $J = 0$, take $M > 0$ and $Q > 0$, and begin at extremality with $Q = M$, $r_+ = M$ and $\Phi_{\mathrm{H}} = Q/r_+ = 1$. Send in a particle with $E > 0$ and $q > 0$. Destruction would require $Q + q > M + E$ and that's just

$$q > E.$$

But the condition (5) implies $E - \Phi_{\mathrm{H}} q \geq 0$ so

$$E \geq q.$$

Our two requirements are

$$q > E \qquad \text{and} \qquad E \geq q$$

while there is no overlap. The particle with the right forged passport is the particle turned away at the door. This is not teleology (the philosophical idea that things exist, happen, or develop because of an inherent purpose, goal, or final end, rather than simply from prior causes). The black hole does not calculate its proposed final charge and make a decision. The Coulomb field and the future directedness of the trajectory enforce the same inequality locally.

4.2 Attempt II: Overspin an extremal Kerr hole

Now let $Q = 0$ and begin with $J = M^2$, $a = M$ and $\Omega_{\mathrm{H}} = 1/2M$. After absorbing (E, L), the hole would be overspun if $J + L > (M + E)^2$ and that's

$$L > 2ME + E^2.$$

Yet capture requires $E - \Omega_{\mathrm{H}} L \geq 0$ so

$$L \leq 2ME.$$

Again, there is no overlap

$$L \leq 2ME \qquad \text{and} \qquad L > 2ME + E^2.$$

The extra E^2 makes the failure even stronger. The body cannot overspin the hole even by saturating the horizon toll. The Reissner–Nordström and Kerr calculations look like separate tricks. They are shadows of one simple geometric fact. We can solve $\mathcal{F} = 0$ for the positive extremal mass

$$M_{\mathrm{ext}}(J, Q) = \left(\frac{Q^2 + \sqrt{Q^4 + 4J^2}}{2} \right)^{1/2}.$$

Since stationary Kerr–Newman black holes satisfy $M \geq M_{\mathrm{ext}}(J, Q)$, M_{ext} is the least mass a Kerr–Newman horizon can have while carrying the specified J and Q. We can differentiate the extremality equation along the extremal surface

$$M^4 - M^2 Q^2 - J^2 = 0.$$

$$(4M^3 - 2MQ^2)\mathrm{d}M - 2M^2 Q\,\mathrm{d}Q - 2J\,\mathrm{d}J = 0,$$

$$\mathrm{d}M_{\mathrm{ext}} = \frac{J}{M(2M^2 - Q^2)}\,\mathrm{d}J + \frac{QM}{2M^2 - Q^2}\,\mathrm{d}Q. \tag{8}$$

At extremality, $r_+ = M$ and $r_+^2 + a^2 = M^2 + a^2 = 2M^2 - Q^2$. Using (2), equation (8) becomes

$$\mathrm{d}M_{\mathrm{ext}} = \Omega_{\mathrm{H}}\,\mathrm{d}J + \Phi_{\mathrm{H}}\,\mathrm{d}Q. \tag{9}$$

Now place (9) beside the admission law (6) to have

$$\delta M \geq \Omega_{\mathrm{H}} \delta J + \Phi_{\mathrm{H}} \delta Q = \delta M_{\mathrm{ext}}.$$

At the extremal boundary, physical matter cannot increase the actual mass by less than the increase in the minimum mass required to support its added charge and angular momentum. At best the perturbation moves tangent to the boundary. More commonly it moves inward, back toward subextremality. This is Wald's two failures in just one sentence! The horizon admission plane supports the black hole region at every extremal Kerr–Newman point. The first law of black hole mechanics is that

$$\delta M = \frac{\kappa}{8\pi} \delta A_{\mathrm{H}} + \Omega_{\mathrm{H}} \delta J + \Phi_{\mathrm{H}} \delta Q. \tag{10}$$

Bardeen, Carter, and Hawking organized the laws of black hole mechanics in 1973, before Hawking radiation made the thermodynamic analogy strikingly literal [7]. In (10), the same

Ω_{H} and Φ_{H} reappear because they are conjugate potentials for the same conserved charges. For a nonextremal hole, $\kappa > 0$. Under the assumptions of the classical area theorem, we've got $\delta A_{\mathrm{H}} \geq 0$, so (10) again yields the admission inequality. Reversible transformations have $\delta A_{\mathrm{H}} = 0$ and exactly saturate the toll while irreversible ones pay extra by increasing the area. There is another useful ledger. We can define the irreducible mass

$$M_{\mathrm{irr}} = \sqrt{\frac{A_{\mathrm{H}}}{16\pi}}.$$

The Christodoulou–Ruffini relation reads [8] that

$$M^2 = \left(M_{\mathrm{irr}} + \frac{Q^2}{4M_{\mathrm{irr}}} \right)^2 + \frac{J^2}{4M_{\mathrm{irr}}^2}. \tag{11}$$

Classical extraction can reduce the rotational or electromagnetic part of the mass, but the area theorem prevents M_{irr} from decreasing. Extremality is the minimum of (11) at fixed J, Q. The reducible energy can be exhausted but it cannot be extracted twice. Equations (5), (7), (9), and (10) are four readings of the same horizon generator and the same conserved charges. At exact extremality, the tangent plane argument is merciless. So begin just inside the black hole region. A near extremal hole possesses a small but nonzero gap, and its horizon potential is slightly cheaper than its extremal value. Perhaps the projectile can enter at today's price and leave the hole unable to afford tomorrow's charges. This is not a silly idea. Hubeny showed in 1999 that, when backreaction is neglected, a near extremal Reissner–Nordström hole admits an apparent overcharging window [9]. A decade later, Jacobson and Sotiriou found the rotating counterpart [10]. The important word is not *apparent*. It is *window*. We should calculate its width before deciding whether the terms outside the test body approximation are relevant. Let $M = 1$ temporarily and I shall parameterize a near-extremal Reissner–Nordström hole by $Q = \sqrt{1 - \epsilon^2}$, $r_+ = 1 + \epsilon$ and $0 < \epsilon \ll 1$. Then

$$\Phi_{\mathrm{H}} = \frac{Q}{r_+} = \sqrt{\frac{1 - \epsilon}{1 + \epsilon}} = 1 - \epsilon + \frac{\epsilon^2}{2} + O(\epsilon^3).$$

The horizon crossing inequality gives the necessary lower bound

$$E \geq E_{\min} = q\Phi_{\mathrm{H}} = q\left(1 - \epsilon + \frac{\epsilon^2}{2} + \cdots\right).$$

Overcharging requires $Q + q > 1 + E$ or

$$E < E_{\max} = q - \left(1 - \sqrt{1 - \epsilon^2}\right) = q - \frac{\epsilon^2}{2} + O(\epsilon^4).$$

$$E_{\max} - E_{\min} = q\epsilon - \frac{\epsilon^2}{2} - \frac{q\epsilon^2}{2} + O(\epsilon^3 q, \epsilon^4).$$

If $q = \alpha\epsilon$, the leading width is

$$E_{\max} - E_{\min} = \left(\alpha - \frac{1}{2}\right)\epsilon^2 + O(\epsilon^3).$$

For $\alpha > 1/2$, the two inequalities overlap at test particle order. A particle can appear both capturable and destructive. For radial motion, the charged particle equation can be written as

$$\mu^2 \dot{r}^2 = \left(E - \frac{qQ}{r}\right)^2 - \mu^2 \left(1 - \frac{2M}{r} + \frac{Q^2}{r^2}\right)$$

and a complete test particle analysis must also check that no turning point blocks the route from the chosen initial position. Hubeny exhibited compatible parameter ranges. The horizon inequality alone is necessary, not sufficient and the complete effective potential matters. But none of that changes the crucial scaling, the apparent victory is only $O(\epsilon^2)$. The Kerr version tells the same story. Again let $M = 1$, and this time we've got $J = a = 1 - 2\epsilon^2$ with $0 < \epsilon \ll 1$. Then

$$r_+ = 1 + \sqrt{1 - a^2} = 1 + 2\epsilon + O(\epsilon^3)$$

and, using $r_+^2 + a^2 = 2r_+$ for Kerr with $M = 1$, we have

$$\frac{1}{\Omega_{\mathrm{H}}} = \frac{2r_+}{a} = 2 + 4\epsilon + O(\epsilon^2).$$

The capture condition gives

$$L \leq L_{\max} = \frac{E}{\Omega_{\mathrm{H}}} = 2E + 4\epsilon E + O(\epsilon^2 E).$$

The final state is overspun if $J + L > (1 + E)^2$ or

$$L > L_{\min} = 2E + E^2 + 2\epsilon^2.$$

An overlap requires, at leading nontrivial order, that

$$E^2 - 4\epsilon E + 2\epsilon^2 < 0.$$

$$(2 - \sqrt{2})\epsilon < E < (2 + \sqrt{2})\epsilon.$$

Once again $E = O(\epsilon)$, $L = O(\epsilon)$, and the width separating success from failure is $O(\epsilon^2)$. There is a beautiful irony here. The exactly extremal hole is easy to protect because first order decides the case. The slightly nonextremal hole looks vulnerable because a first order calculation produces a second order verdict. That sentence should make us suspicious. Let λ label a one parameter family of physical configurations

$$\begin{aligned} M(\lambda) &= M + \lambda\,\delta M + \frac{\lambda^2}{2}\delta^2 M + O(\lambda^3), \\ J(\lambda) &= J + \lambda\,\delta J + \frac{\lambda^2}{2}\delta^2 J + O(\lambda^3), \\ Q(\lambda) &= Q + \lambda\,\delta Q + \frac{\lambda^2}{2}\delta^2 Q + O(\lambda^3). \end{aligned} \tag{12}$$

The extremality diagnostic is

$$f(\lambda) = M(\lambda)^2 - Q(\lambda)^2 - \frac{J(\lambda)^2}{M(\lambda)^2}.$$

For the initial hole, $f(0) = M^2\epsilon^2$. The dangerous process is tuned to make the first order horizon flux as small as possible, with

$$\delta M - \Omega_{\mathrm{H}}\delta J - \Phi_{\mathrm{H}}\delta Q = 0. \tag{13}$$

If $\lambda \sim \epsilon$, then $f(0) \sim \epsilon^2$, $\lambda\,\delta f \sim \epsilon^2$ and $\lambda^2\delta^2 f \sim \epsilon^2$. The initial gap, the claimed violation, and the second order correction are parametrically equal. Now note what a strict test particle model omits (oh boy): electromagnetic and gravitational self force, the body's electromagnetic self energy, stresses and finite size required to store charge or spin, gravitational and electromagnetic radiation, changes in the capture threshold caused by the perturbed geometry and even nonlinear interaction energy in the combined body–field–black hole system. Each is small in an absolute sense. That is irrelevant. The verdict is also small. If $q \sim \epsilon M$, then $q^2/M \sim \epsilon^2 M$, exactly the scale of the Hubeny window. If $E \sim \epsilon M$, then $E^2 \sim \epsilon^2 M^2$, exactly the scale of the Jacobson–Sotiriou window. Neglecting a term because it is small is legitimate only when the question being answered is larger. Here the alleged naked singularity fits inside the error bar. One could try to calculate every omitted effect separately. That program is ferocious for general Kerr–Newman perturbations. Sorce and Wald took a more powerful route in 2017: instead of itemizing the entire bill, they derived a variational identity that constrains the total [11].

5 Magic

Consider Einstein–Maxwell theory with fields $\phi = (g_{ab}, A_a)$ and Lagrangian four form

$$\boldsymbol{L} = \frac{1}{16\pi}\left(R - F_{ab}F^{ab}\right)\boldsymbol{\epsilon}.$$

For a general diffeomorphism covariant theory, the first variation has the form

$$\delta\boldsymbol{L} = \boldsymbol{E}(\phi)\cdot\delta\phi + \mathrm{d}\boldsymbol{\theta}(\phi, \delta\phi),$$

where $\boldsymbol{E} = 0$ are the field equations and $\boldsymbol{\theta}$ is the symplectic potential three form. Antisymmetrizing a second variation defines the symplectic current

$$\boldsymbol{\omega}(\phi; \delta_1\phi, \delta_2\phi) = \delta_1\boldsymbol{\theta}(\phi, \delta_2\phi) - \delta_2\boldsymbol{\theta}(\phi, \delta_1\phi).$$

For a vector field X^a, the Noether current is

$$\boldsymbol{J}_X = \boldsymbol{\theta}(\phi, \mathcal{L}_X\phi) - X\cdot\boldsymbol{L}.$$

and it decomposes as

$$\boldsymbol{J}_X = \boldsymbol{C}_X + \mathrm{d}\boldsymbol{Q}_X$$

where $\boldsymbol{C}_X$ encodes the constraints and $\boldsymbol{Q}_X$ is the Noether charge two form. Combining these equations and integrating over a hypersurface converts bulk dynamics into boundary charges plus flux. This is the covariant phase space framework developed by Iyer and Wald [12]. Choose $X^a = \xi^a = t^a + \Omega_{\mathrm{H}}\varphi^a$, the background horizon generator. At first order, the boundary term at infinity is $\delta M - \Omega_{\mathrm{H}}\delta J$, while the electromagnetic horizon contribution gives $\Phi_{\mathrm{H}}\delta Q$. The resulting flux identity yields

$$\delta M - \Omega_{\mathrm{H}}\delta J - \Phi_{\mathrm{H}}\delta Q = \text{nonnegative horizon flux}$$

under the null energy condition. Thus the admission ledger was already a Noether identity! At second order, the symplectic current defines the canonical energy

$$\mathcal{E}_\Sigma(\phi;\delta\phi) = \int_\Sigma \omega(\phi;\delta\phi,\mathcal{L}_\xi\delta\phi).$$

This quantity has the quadratic gravitational and electromagnetic perturbation energy, including the order at which the proposed violation lives. For an optimal first order process satisfying (13), Sorce and Wald compare the physical perturbation with a Kerr–Newman family having the same first order $(\delta M, \delta J, \delta Q)$. Positivity of the relevant fluxes and late time settling then imply

$$\delta^2 M - \Omega_{\mathrm{H}}\delta^2 J - \Phi_{\mathrm{H}}\delta^2 Q \geq -\frac{\kappa}{8\pi}\,\delta^2 A^{\mathrm{KN}}. \tag{14}$$

Here $\delta^2 A^{\mathrm{KN}}$ is the second variation of the horizon area of the comparison Kerr–Newman family. The right hand side is not an arbitrary correction appended to save censorship. It is fixed by the curvature of the constant area surface in parameter space. Return to

$$f(\lambda) = M(\lambda)^2 - Q(\lambda)^2 - \frac{J(\lambda)^2}{M(\lambda)^2}$$

and the near extremality parameter ϵ in (1). Combining the first order optimum (13) with the second order inequality (14), Sorce and Wald obtain, to the controlled order, that

$$f(\lambda) \geq \left[M\epsilon + \frac{(J^2 - M^4)Q\,\delta Q - 2JM^2\,\delta J}{M(M^4 + J^2)}\,\lambda\right]^2 + O(\lambda^3, \epsilon^3, \epsilon^2\lambda, \epsilon\lambda^2).$$

The long calculation ends with the shortest possible verdict that $f(\lambda) \geq 0$ through second order. That square is the spine reaching its destination. The geometric ledger began with f. The admission ledger selected the optimal first order perturbation. The perturbative ledger supplied the curvature correction. When all three are balanced at the order on which the answer depends, the route to $f < 0$ closes. The rescue is not a large effect overwhelming a small attack. It is a small effect defeating an equally small claim. Back reaction is not the deus ex machina (god from the machine). It is the receipt. Precision matters most when the conclusion sounds grand. The Sorce–Wald analysis concerns perturbations of extremal or nearly extremal Kerr–Newman black holes in four dimensional classical Einstein–Maxwell theory. Its argument requires, in particular, a well defined asymptotic charges M, J, Q, suitable regularity and a controlled one-parameter perturbation, the null energy condition for the non electromagnetic matter, an optimal process analysis at first order and, for the near extremal second order argument, appropriate late time settling or linear stability of the background.

Within that domain, generalized Hubeny like accretion processes cannot overcharge or overspin the hole through the relevant order. This is powerful evidence for weak cosmic censorship in a sharp manner. It is not a universal proof of weak cosmic censorship, and it says nothing automatic about arbitrary modified gravities, large non perturbative collisions, quantum stress tensors, exotic boundary conditions, or matter violating the hypothesis used to obtain the flux inequality. If the gravitational theory changes, the first task is not to reuse $M^2 \geq a^2 + Q^2$ ceremonially. One must derive the theory's stationary black hole family, its physical conserved charges, its extremality surface, its horizon potentials and flux law and the correct second order variational inequality. Only then is there a destruction experiment to discuss.

6 Tempting shortcuts

The traditional third law of black hole mechanics says, roughly, that a regular finite process cannot drive a nonextremal horizon to zero surface gravity. It offers excellent intuition: if $\kappa = 0$ is unattainable, crossing $\kappa = 0$ sounds hopeless. But it is not the proof we need. Kehle and Unger constructed regular, spherically symmetric solutions of the Einstein–Maxwell–charged-scalar system that become exactly extremal Reissner–Nordström after finite advanced time [13]. Their result shows that broad unattainability statements depend delicately on the matter model and hypotheses. This does not produce a superextremal naked singularity. Reaching the boundary and crossing it are different mathematical questions. The second order censorship argument is stronger for our purpose because it controls the sign of the extremality function rather than invoking a general slogan about κ. For the exact Kerr–Newman metric, $f < 0$ removes the real roots of Δ while leaving $\Sigma = 0$, so the ring singularity is naked. In a dynamical experiment, however, total asymptotic charges alone do not prove that the final spacetime is a single stationary electrovacuum Kerr–Newman geometry. The system might radiate, retain exterior matter, fragment, fail to settle, or form a horizon whose relation to the total charges is not captured by a one body stationary ansatz. Thus (3) is the right target under a final state assumption. A genuine censorship violation is a statement about the causal structure of the full development. This is also why apparent horizons and event horizons must not be conflated. Marginally trapped surfaces are quasi local and numerically useful. Their temporary absence is not equivalent to the absence of the global boundary $\partial J^-(\mathscr{I}^+)$.

Perhaps exotic or quantum matter can simply pay negative energy? Quantum fields can violate classical pointwise energy conditions. That fact does not license us to insert a negative energy pebble into the classical test body equations and declare the horizon unemployed. Semiclassical gravity changes the stress tensor, fluctuations, entropy balance, evaporation, and sometimes the appropriate notion of horizon. A quantum experiment must keep its own ledger consistently. Likewise, the question is not whether $M^2 \geq a^2 + Q^2$ survives in every theory. It plainly need not be the correct bound outside Einstein–Maxwell theory. The interesting question is whether the theory's own extremal energy bound can be crossed by a regular process allowed by that same theory. By now the resistance of the horizon can sound conspiratorial. A dangerous charged particle is repelled. A dangerous high L/E orbit is scattered. A dangerous bosonic frequency is superradiant. General matter obeys a flux inequality. The near extremal loophole opens only at the order where backreaction closes it. *Is a committee of cosmic censors hiding just behind r_+?* No. There is one local structure $\xi^a = t^a + \Omega_{\mathrm{H}}\varphi^a$ and one associated energy $\mathcal{E}_\xi = E - \Omega_{\mathrm{H}}L - \Phi_{\mathrm{H}}q$. Future directed horizon crossing requires $\mathcal{E}_\xi \geq 0$. The same generator defines the thermodynamic potentials. Those potentials are the derivatives of the extremal energy. The symplectic flux generated by ξ^a becomes canonical energy at second order. The global looking protection is assembled from local propagation, conserved charges, and the nonlinear field equations. The horizon does not know the future. The equations refuse to sell the required J and Q for less than their support energy.

A recent 2026 preprint proposes a complementary geometric version of this idea: under null convergence and a genericity condition, injected matter focuses the initial horizon generators so that a horizon cross section becomes a closed trapped surface and candidate final geometries unable to contain that surface are then excluded [14]. Because this approach is formulated through trapped surfaces rather than only asymptotic extremality bounds, it suggests that

the censoring mechanism may be more local and geometric than the Kerr–Newman charge ledger alone reveals. As a new result, it should be read with the usual preprint caution (I mean, that's science). Its direction, however, is telling that the field equations leave geometric evidence of the horizon before the final stationary accountant arrives.

7 Failure?!

We can now state the attempted operation without euphemism (an indirect word or expression that you use instead of a more direct one when you are talking about something that is unpleasant or embarrassing, the use of such expressions). So we begin with a Kerr–Newman black hole

$$\mathcal{F}(M, J, Q) = M^4 - M^2Q^2 - J^2 \geq 0.$$

Send matter carrying energy, angular momentum, and charge into it. To destroy the stationary horizon, the final charges must satisfy

$$\mathcal{F}(M_{\mathrm{f}}, J_{\mathrm{f}}, Q_{\mathrm{f}}) < 0.$$

But the projectile must cross the existing horizon, and crossing imposes

$$\delta M - \Omega_{\mathrm{H}}\delta J - \Phi_{\mathrm{H}}\delta Q \geq 0.$$

At exact extremality, the right hand side

$$\Omega_{\mathrm{H}}\delta J + \Phi_{\mathrm{H}}\delta Q$$

is exactly the first variation of the minimum mass $M_{\mathrm{ext}}(J, Q)$. The best admissible perturbation is tangent to the extremal surface. Overcharging extremal Reissner–Nordström demands $q > E$, while capture demands $E \geq q$. Overspinning extremal Kerr demands $L > 2ME + E^2$, while capture demands $L \leq 2ME$. The perfect crimes fail before the projectile crosses the door. Near extremality, the admission price drops just enough to create a test-body window. That is the intellectually honest moment of danger. But the window is $O(\epsilon^2)$, while the projectile itself is $O(\epsilon)$. Self energy, self force, radiation, finite size, and nonlinear backreaction are therefore not optional embellishments (a decorative detail added to make something more attractive or interesting), they enter at the order of the verdict. The second order variational inequality completes the accounting and reorganizes the final extremality function into the nonnegative square.

So the best answer is not merely that "cosmic censorship wins." The deeper answer is a chain of identities. The horizon generator implies that $E - \Omega_{\mathrm{H}}L - \Phi_{\mathrm{H}}q \geq 0$, the extremal surface implies that $\mathrm{d}M_{\mathrm{ext}} = \Omega_{\mathrm{H}}\,\mathrm{d}J + \Phi_{\mathrm{H}}\,\mathrm{d}Q$ and the canonical energy implies that $f(\lambda) \geq (\text{real quantity})^2 + O(3)$. Charge and rotation do weaken a Kerr–Newman horizon. They reduce $r_+ - r_- = 2M\epsilon$ and lower $\kappa = M\epsilon/r_+^2 + a^2$. One may approach the edge until the throat becomes long and the surface gravity becomes tiny. What one cannot do, within the classical perturbative Einstein–Maxwell experiment described above, is deliver the dangerous charges more cheaply than the field equations allow. The black hole survives for a reason more satisfying than invincibility. It survives because every successful looking attack has quietly borrowed from a term it has not yet paid. Exact extremality catches the fraud at first order. Near extremality waits for the second order to mail.

And that is how to fail at destroying a black hole.

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